MARIO ALEJANDRO MARROQUÍN

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EDUCATION

Complejo Educativo "Thomas Jefferson" El Salvador
Technical Baccalaureate in Technology Infrastructure and IT Services *Expected 2026*

- Straight-10 grades and first place in class every year. Coursework: networks, operating systems, information security. Conducted an ISO 27001 audit of a savings and credit cooperative.

EXPERIENCE

Accounting Assistant, Process Automation Feb 2026 – Present
El Salvador

- Built Partidas DTE, a desktop app that reads identifiers from accounting entries (PDF) and merges the matching electronic invoices into one file, cutting a task that took a full afternoon (~4 hours) to under 1 minute (over 99% faster), recurring monthly across multiple clients.
- Built DTE Joiner, a second tool that reads the purchases ledger (Excel) and assembles the matching invoice PDFs in ledger order, with resumable progress, missing-invoice alerts and an audit log; similar reduction (~4 hours to under 1 minute).
- Developed Excel/VBA macros that generate payroll sheets and journal entries with automatic balance checks, saving an estimated 30 to 40% of payroll time; automated data validation with Python.

Math Tutor, Community Service 200 hours
El Salvador

- Taught mathematics classes, including to grades above my own.

RESEARCH & PROJECTS

CryptoRisk github.com/mariomarroquin3/cryptorisk

- Compared 20+ volatility and tail-risk models (GARCH family, HAR, CAViaR, MS-GARCH, LSTM) for 1-day VaR and ES on BTC and ETH with a 500-day walk-forward protocol; evaluated with Kupiec, Christoffersen, Diebold-Mariano and MCS tests.
- Deployed a live dashboard (Next.js, React) and wrote the methodology in LaTeX.

Entropy Arena github.com/mariomarroquin3/entropy-arena

- Designed a framework to compare randomness sources for cryptographic key generation (BIP39/32/48/67); recovered the MT19937 state in 0.02 s and RANDU in 0.26 s.

Violence, Institutions and Growth github.com/mariomarroquin3/data_analyzer

- Built a reproducible two-way fixed-effects pipeline on 8 countries (2000–2024) with small-cluster inference (wild cluster bootstrap, CR2, Driscoll-Kraay) and ML triangulation.

RetinoVisionAI · Financial data VAE · IoT

- Trained an EfficientNet-B3 classifier for diabetic retinopathy with Grad-CAM (PyTorch, Flask); entered in the EUREKA challenge.
- Built a Variational Autoencoder (PyTorch, Streamlit) that generates synthetic financial transactions for software testing without exposing real data.
- Implemented RFID access control (ESP32, Firebase, React) and Zabbix host monitoring on Ubuntu (team project).

HONORS, LEADERSHIP & PROGRAMS

- Winner, first National AI Hackathon (2025): coffee quality classification with a Random Forest model.
- Multiple wins in district and departmental public speaking contests. Mathematics olympiads, two consecutive years.
- Member, Cohere Labs Open Science Community and Google Developer Program; participant, ML Summer School 2026.
- In progress: GCI (Matsuo Lab, University of Tokyo), CUBO+, CUBO_ai, Stanford Probability for AI (admitted, fall 2026).

CERTIFICATIONS

- AWS: ML Engineer Associate Curriculum Overview; Introduction to Machine Learning; SimuLearn Cloud Fundamentals (2026).
- Stanford Code in Place (2026) · Cisco Introduction to Data Science (2026) · Udacity AWS AI Practitioner Challenge (2026).
- Platzi: Linear Algebra for ML; Math Functions for Data Science and AI; Math Language and Notation; Math Fundamentals (2026).
- Google ML Crash Course badges: Linear Regression, Logistic Regression · Daxus LATAM: Power BI + AI Career Accelerator, 8 h (2026) · Direct English, Intermediate (2024).

SKILLS & LANGUAGES

Programming: Python, SQL, LaTeX. Working knowledge with AI assistance: TypeScript, JavaScript/React, Java, VBA; learning R.
ML & Data: PyTorch, scikit-learn, statsmodels, arch, Pandas, NumPy, Streamlit, Flask, Next.js.
Tools & Infrastructure: Git/GitHub, Power BI, Firebase, Zabbix, Ubuntu, ESP32. AI-assisted engineering: architecture and feature design with AI agents and MCP.
Languages: Spanish (native); English (intermediate certificate 2024, advanced level in preparation).